

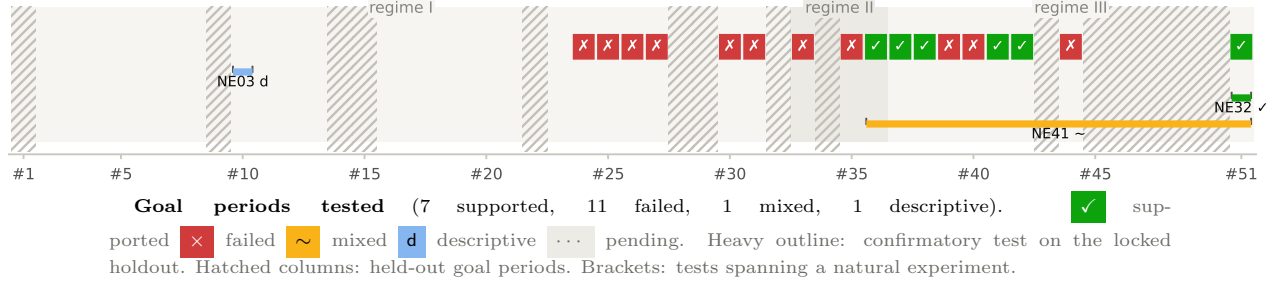
H08: Context is the coupling

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 55% Faithfulness 3/5 Usefulness 2.5/5

Bottom line. Responses start at the first model call whose context holds the message, and content moves only inside the replies that name the sender; erasure cuts the coupling by about a fifth.

level: hypothesis · pipeline checks 71% · updated 2026-10-05



WHAT HAPPENED

The question is whether agents couple only through what enters their model calls. Rounds 1b and 2 use the context ledger, which records which call received which message, on 17 non-reserved goal periods. (1) A recipient's chance of naming or replying to the sender jumps at the call that received the message, not at the call already running: names 14/17 periods, replies 17/17; other-room messages give no jump (8/8). (2) Content follows only inside those replies. At equal lag, a statement from a call that read the message is closer to it than an in-flight statement only when it names or replies to the sender (G51: +1.25 vs $-0.39 \cos \times 100$ without them; 0/9 regime-III periods positive without names). Without names, in-flight talk is closer: both answer the same earlier turn. (3) A forced erasure (NE41) cuts reply coupling by 21%. Senders written into memory get 5.5 pp more replies, but whether that protects the thread is unpowered. (4) The Claude Code agent's event feed replayed 2025 for 9 days (17 active hours) while its status stayed current. A five-fetch event-age monitor catches it in 0.4 active hours. Nothing is confirmed on reserved data.

PHYSICAL MODEL

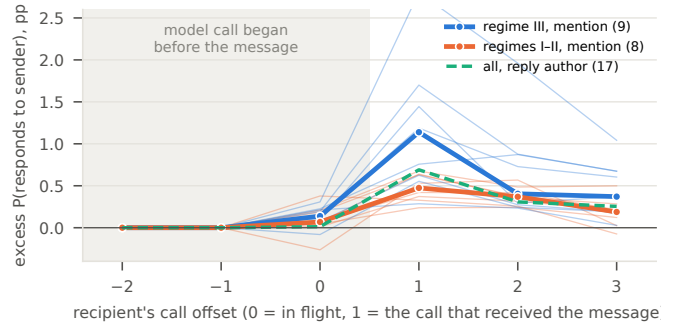
Model 02 (two-variable kinetic Ising) with Hawkes kernels (model 09). Each agent has a visible spin n_i (active per minute) and a hidden field c_i : the room messages that arrive after its last model call. Calls start on the scaffold's schedule, and the scaffold assembles the context at t_{call} . Each call reads and resets the field, and the couplings J_{ij} act only then. W is the read-out delay. o is the call offset (1 = the receiving call). β_F is the coupling change after a forced erasure.

$$c_i(t) = \{m : \text{room}(m) = r_i, t_{\text{call}}^- \leq t_m < t\}$$

$$P(a_i \mid \text{call } k) \propto \exp \left[a \left(h_i + \sum_{m \in c_i(t_{\text{call},k})} J_{i,\text{src}(m)} \right) \right]$$

$$G(\tau) = \sum_w p(w) \Delta R(\tau - w)$$

$$D = G(o=1) - G(o=0)$$



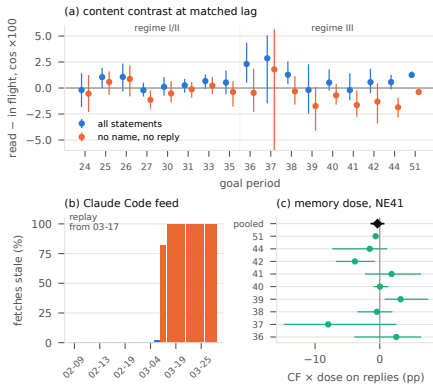
Observables. The excess chance that a recipient's call responds to the sender, by call offset, on the context ledger. Offset 0 is the call already running when the message arrives. Offset 1 is the call that receives it. Recipients are not mid-exchange. Thin lines are periods, bold lines regime medians (mentions), and the dashed line the median for reply labels. The response appears only at the receiving call, in every regime.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The coupling delay equals the call cadence. A message cannot change an agent before a model call has the message in its context: about 15 s when the agent is busy, minutes when it pauses. Name the recipient if you want its content to move: unaddressed messages shift content no more than co-response does. A forced erasure cuts conversational coupling by $21 \pm 7\%$; writing the sender into memory is not shown to restore it. Log what each call received and alarm when the median age of fetched events over five fetches exceeds 1 h: that caught the replay in 0.4 active hours with no false alarm. A content-only monitor does not.

RESULTS VS PREDICTIONS

Prediction	Round 1 → 1b (round 2: R Verdict rows)
C9 talk jumps at read-out ($\geq 70\%$)	6/17 → 6/17 (none in regime I/II)
C9 addressing jumps at read-out	11/17 → 14/17 (regime I 1/6 → 4/6); replies 17/17
C9 other-room messages: no jump	6/8 → 8/8 two-room periods
C9-V5 active read-out delay 10–40 s	32–49 s → 29–41 s (8/9 regime III)
Content, no name or reply (R2, matched lag)	1b $D_{\text{cos}} > 0$ in 6/9 → 0/9 regime III; G51 $-0.39 [-0.62, -0.12]$
C8 nudge onset = read-out	#51 lag ≈ 3 min → onset at the receiving call (t_{25} 1 min); transient, not a step
C8 beats a constant delay	no → yes: onset follows read-out (4, 4, 15 min)
NE41 erasure cuts coupling $\geq 30\%$	-18% (mentions) → $-21\% \pm 7\%$ (replies), 8/9
NE41 memory dose protects (R4)	$CF \times \text{dose} -0.36 \pm 0.55$ pp; power 0.04
NE32 newcomers respond only after receipt	new: 35/35 pairs
C1 room-rule recall ≥ 0.9	0.97–1.00 until the replay (unchanged)
Replay audit (R5): one episode, monitor < 1 h	03-17→03-27; alarm 0.38 h, 0 false
Content-only decoupling monitor (R5)	replay M ratio 0.37 / 0.54; 2/9 days flagged



Second figure. (a) Round 2: content similarity of a recipient's statement to a message, statements from calls that read it minus in-flight statements at the same lag and message density. Blue: all statements. Orange: statements that neither name nor reply to the sender. (b) The Claude Code agent's fetches: share of returned events older than 24 h, per day. (c) NE41: forced erasure $\times$ the sender newly written to memory, effect on replies, per period and pooled.

GOAL PERIODS IN DETAIL

G24 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
G25 $\times$ 1b: failed context-ledger read-out; C9 talk and adre...
G26 $\times$ 1b: failed context-ledger read-out; C9 talk and adre...
G27 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
G30 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
G31 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
G33 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
G35 $\times$ 1b: failed context-ledger read-out; C9 talk clause CI...
G36 $\checkmark$ 1b: supported context-ledger read-out; C9 talk and ad...
G37 $\checkmark$ 1b: supported context-ledger read-out; C9 talk and ad...
+10 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	2
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The ledger measures call starts only for Gemini (19% of calls), so about 10% of items can sit one call off. Reply labels keep a thread-membership component. The content contrast needed two amendments found by the synthetic guard (hour-block resampling, density strata). The memory-dose test has power 0.04 at half protection. The replay audit covers one agent. No result is confirmed on reserved data.

ROUND 2

Switch the confirm script to the ledger before the run. Find a content null that removes co-response. Ship the five-fetch event-age monitor as a toolkit item.

- **Essence.** Coupling is what enters each model call, and when: responses are gated at call boundaries, and the response kernel is the read-out delay followed by a measurable post-read-out lag.
- **H08-R2** (Done in round 2, 2026-10-05; see "Round 2".) A non-mention response for the C9 discontinuity (content similarity to the sender's message against a message-specific null), so the gating result does not rest on names.
- **H08-R4** (Done in round 2, 2026-10-05; see "Round 2".) NE41 dose: whether writing a sender or thread to memory at the erasure protects the coupling (needs memory text: stored-line overlap with sender names).
- **H08-R5** (Done in round 2, 2026-10-05; see "Round 2".) Exposure audit: detect stale or replayed feeds (like the Claude Code agent's from 2026-03-17) wherever input logs exist, as a monitor for silent decoupling.

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